Math-M-Addicts
MINDS THAT COUNT

SP250 Homework 6

1. There are 5 boys and 5 girls in a chess club.

   a) How many ways are there to pick 5 students and send them to a competition?
   b) How many ways are there to do so if there must be at least one boy and one girl on the team?
   c) How many ways are there to do so if there must be at least two boys and at least two girls on the team?
   d) Anya, Ben, Cecily, David and Ellie were picked. How many ways are there to assign them to play boards 1,2,3,4,5 (each student must play exactly at one board)
   e) How many ways are there to do so if Ben and David can't be sitting next to each other? (in other words, they can't play boards 1 and 2, or 2 and 3, etc.).

Solution:

a) Choosing 5 students out of 10 is just $\binom{10}{5} = \frac{10!}{5!5!} = 252$
b) Two of the ways of picking 5 students are not allowed: all girls or all boys. So we have $\binom{10}{5} - 2 = 250$
c) We can have either 2 girls and 3 boys or 3 girls and 2 boys. For the case of 2 girls and 3 boys, there are $\binom{5}{2} = 10$ ways to pick the girls and $\binom{5}{3} = 10$ ways to pick the boys, for a total of $10 \cdot 10 = 100$ ways. Similarly, there are also 100 ways to pick 3 girls and 2 boys, so the overall total number of ways for both cases is 200.
d) $5! = 120$
e) Let's first count the number of arrangements where Ben and David are sitting next to each other. Since they are sitting next to each other, we can consider them as one group, with the other three students each being a group of one. There are 4! ways to arrange the 4 groups. Within the group of Ben and David, we can also switch the order between Ben and David, so the total number of arrangements where Ben and David are next to each other is $2 \cdot 4! = 48$. Thus, the total number of arrangements where Ben and David are not sitting next to each other is $5! - 2 \cdot 4! = 120 - 48 = 72$.

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2. How many 3-digit integers are odd and consist of 3 distinct digits?

Solution 1:

Let's consider two cases based on whether the digit zero is used or not.

Case 1: The digit zero is not used

There are 5 ways to pick the rightmost digit since the number is odd and has to end in one of the 5 odd digits; 8 ways to pick the middle digit since there are 9 nonzero digits and we can't reuse the rightmost digit; and 7 ways to pick the leftmost digit since we can't reuse the middle and rightmost digits. This gives a total of $7 \cdot 8 \cdot 5 = 280$ ways.

Case 2: The digit zero is used

The digit zero can only appear as the middle digit. Hence, there are 5 ways of picking the rightmost digit since it has to be one of the odd digits; 1 way of picking the middle digit since 0 is the only choice; and 8 ways of picking the leftmost digit since we can't reuse the rightmost digit. This gives a total of $8 \cdot 1 \cdot 5 = 40$ ways.

Combining the two cases, we get a total of $280 + 40 = 320$ such 3-digit positive integers.

Solution 2:

There are 5 ways to pick the rightmost digit since the number is odd and has to end in one of the 5 odd digits; 8 ways to pick the leftmost digit since we can't use the zero digit or the rightmost digit; and 8 ways to pick the middle digit as we can't reuse the rightmost and leftmost digits. This gives $8 \cdot 8 \cdot 5 = 320$ ways to construct such a 3-digit number.

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3. 101 integers are chosen from the set {1,2,..., 300}. Show that the difference between two of them is at most 2.

Solution:

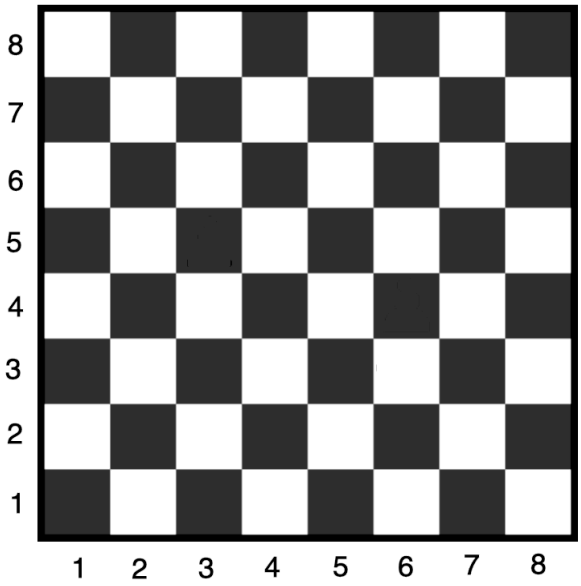
Let's separate the integers in {1, 2, ..., 300} into 100 groups of three consecutive integers {1,2,3}, {4,5,6}, ..., {298,299,300}. By the pigeonhole principle, two of the 101 integers must fall within the same group of three consecutive integers and therefore differ by at most 2.

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4. Is it possible to put 8 rooks on the chessboard, so that no two rooks attack each other and exactly 3 rooks stand on white squares?

Solution:

Let's label the 8 columns and 8 rows of the chessboard with the numbers 1, 2, ..., 8 as shown below.



For any arrangement of 8 rooks on the chessboard, we can add up the row and column numbers of all the rooks. For the 8 rooks to not be attacking each other, every column and every row must contain exactly one rook, so the sum of the row and column numbers for all 8 rooks must be $2 \cdot (1 + 2 + ... + 8)$, which is an even number.

Now, we see that the sum of the row and column numbers of any black square is even and the sum of the row and column numbers of any white square is odd. So, if there are exactly 3 rooks standing on white squares, the sum of all the row and column numbers of the 8 rooks will then be odd. Therefore, it is not possible to place 8 rooks where no two rooks attack each other and exactly 3 rooks stand on white squares.

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5. Prove that the following equation doesn't have integer solutions: $x^2 = 21y + 10$.

Solution:
$21y + 10$ is congruent to 3 mod 7. However, perfect squares can only be 0, 1, 2, or 4 mod 7 as shown in the table below. Hence, this equation cannot have integer solutions.

x	x^2 **(mod 7)**
0	0
1	1
2	4
3	2

4	2
5	4
6	1

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6. Prove that the product of any 100 consecutive positive integers is a multiple of 100! (Hint: let the largest number be n, write out the ratio that you are being asked to show is an integer and stare at it).

Solution:
The product of 100 consecutive positive integers ending in n is $n(n-1)(n-2)\ldots(n-99)$. Dividing by 100! gives

$$\frac{n(n-1)(n-2)\ldots(n-99)}{100!} = \frac{n!}{100!(n-100)!} = \binom{n}{100}$$

$\binom{n}{100}$ is the number of ways to pick 100 items from among n items and so must be an integer. Therefore, $n(n-1)(n-2)\ldots(n-99) = 100! \cdot \binom{n}{100}$ is a multiple of 100!.
